



Short communication

Conformational entropy of self-avoiding polymer chains

Zhihu Wang^a, Mengbo Luo^b, Jianmin Xu^{b,*}^a*Department of Physics, Hangzhou Television University, Hangzhou, 310009, P.R. China*^b*Department of Physics, Hangzhou University, Hangzhou, 310028, P.R. China*

Received 17 March 1997; accepted 25 November 1997

Abstract

The conformational entropy of self-avoiding polymer chains on a tetrahedral lattice is calculated with the exact enumeration method. The conformational entropy linearly varies with the chain length. The parameters governing the length dependence of the conformational entropy appear to be dependent on the temperature and the energy ε , which is the energy of gauche state relative to trans state. The second-order transition temperature T_2 depends mainly on ε . © 1999 Elsevier Science Ltd. All rights reserved.

1. Introduction

The conformational entropy, which can be investigated by a variety of methods [1–5], is an important conformational property associated with the order of the system. When long-range interactions prevail among polymer chains, such as excluded volume, van der Waals potential, etc., one precise method is the exact enumeration of all possible conformations of the polymer chains. Though this method is only able to deal with short chains due to the tremendous number of the conformations for long chains, it has been used by many research groups to study conformational properties of polymer chains.

In the present work, we propose to calculate the conformational entropies of polymer chains with the exact enumeration method. The model used is the self-avoiding chain on a tetrahedral lattice, which can be considered as a model for many real polymers such as polyethylene, poly(oxymethylene), and poly(oxyethylene), etc. [6, 7].

2. Calculation method

The polymer chains are generated on a tetrahedral lattice. There are three possible states for each step, trans, gauche⁺, and gauche[−]. The short-range interaction of the polymer chain is represented by that the gauche state has a higher energy than the trans state, while the long-range interaction is self-avoiding on the lattice.

After having generated all the possible conformations of the self-avoiding polymer chains with N bonds, the conformational entropy is calculated,

$$S = -k_B \sum_i p_i \ln p_i \quad (1)$$

where p_i is the probability of conformation i ,

$$p_i = \frac{\exp(-E_i/k_B T)}{\sum_i \exp(-E_i/k_B T)} \quad (2)$$

The summation is over all the possible conformations, k_B is the Boltzmann constant, T is the temperature, and E_i is the energy of conformation i .

* Corresponding author.

Since the conformational energy E_i is a function of the energy ε and the chain length N , we introduce a dimensionless value $\eta = k_B T / \varepsilon$. Thus, the conformational entropy is a function of η and N .

In this paper, we calculate the conformational entropies of the polymer chains whose gauche state has a positive energy, that is $\eta > 0$, the opposite situation ($\eta < 0$) is reserved for a subsequent article.

3. Results and discussion

The conformational entropies for different η and N are calculated. The η is varied from 0.2 to 1.0 in increments of 0.05, and from 1.0 to 4.0 in increments of 0.1. The chain length is limited to 16 bonds. The conformational entropy is in k_B units hereafter.

Fig. 1 shows the dependence of the conformational entropy on the chain length for various η . It shows that the conformational entropy is linear with the chain length,

$$S = \alpha(N - 2) + \beta \quad (3)$$

α and β are only dependent on the value η .

Though Eq. (3) is derived from the results of short chains, the linear relation of S vs $N-2$ is so good that it seems reasonable to consider that the relation is still valid for long chains. From Fig. 1, it is evident that, for a long chain, the contribution of β in Eq. (3) to the conformational entropy is trivial, so the conformational entropy can be expressed approximately as

$$S = \alpha(N - 2) \quad (4)$$

Thus, the variation of α with η represents that of the conformational entropy S with η .

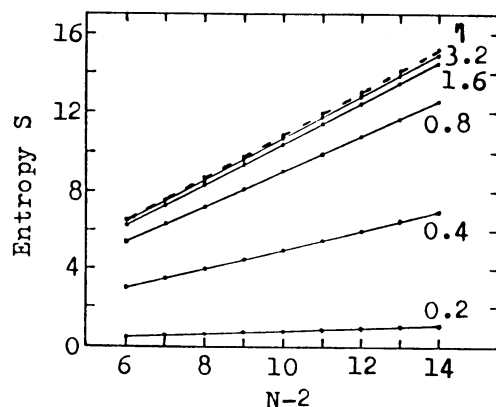


Fig. 1. Plots of the conformational entropy vs chain length for different η . The dashed line is the entropy in the limit of $\eta \rightarrow \infty$.

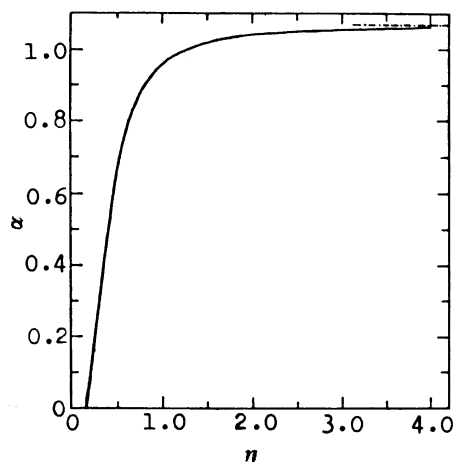


Fig. 2. Plot of the value α vs the value η . The dot-dashed line depicts the value for α in the limit of $\eta \rightarrow \infty$.

Fig. 2 shows the dependence of α on the value η . One can find that the conformational entropy decreases steadily as the value η decreases. It indicates that, for a given polymer chain whose energy ε is fixed, the conformational entropy will decrease as the temperature decreases. This can be explained by that the conformations of the polymer chain rearrange when the temperature decreases. At high temperature, the chains don't prefer any special conformation, but at low temperature, they begin to prefer some low energy conformation. While at the same temperature, the conformational entropy increases with the increment of the chain's flexibility represented by the energy ε for equal chain length.

The value of α in the region $\eta = 0.5-4.0$ can be expressed approximately as

$$\alpha = 0.04942 + 7.338\tau - 26.87\tau^2 + 54.17\tau^3 - 54.93\tau^4 + 21.69\tau^5 \quad (5)$$

where $\tau = \exp(-\varepsilon/k_B T) = \exp(-\eta^{-1})$. And in the region $\eta < 0.5$, the value of α decreases rapidly as η decreases, the plot is nearly linear, α can be expressed as

$$\alpha = 2.104\eta - 0.3431 \quad (6)$$

According to the theory of Gibbs–Dimarzio [8], one can get the second-order transition temperature T_2 where the entropy becomes zero, for a self-avoiding polymer chain on a tetrahedral lattice it is

$$k_B T_2 / \varepsilon \approx 0.16 \quad (7)$$

The temperature T_2 depends mainly on the value ε . It indicates that the more flexible the polymer chain is, the lower the second-order transition temperature is.

In summary, the conformational entropy of a self-avoiding polymer chain on a tetrahedral lattice is linear with chain length, and is also dependent on the temperature and the energy ε . The second-order transition temperature is determined chiefly by the energy ε .

Acknowledgement

Project supported by the Natural Science Foundation of Zhejiang Province, China.

References

- [1] Flory PJ. *Statistical Mechanics of Chain Molecules*. New York: Interscience, 1969.
- [2] Domb C, Hioe FT. *J Chem Phys* 1969;51:1915.
- [3] Meriovitch H. *Macromolecules* 1983;16:249.
- [4] Mazur J, Mccrackin FL. *J Chem Phys* 1968;49:648.
- [5] Collet O, Premilat S. *Macromolecules* 1993;26:6076.
- [6] Winnik MA, Rigby D, Stepto RFT, Lemaire B. *Macromolecules* 1980;13:699.
- [7] Geny F, Monnerie L. *J Polym Sci* 1979;17:131.
- [8] Gibbs JH, Dimarzio EH. *J Chem Phys* 1958;28:373.